AI Market Twin — Admin Console UX/UI Design Spec

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1. Admin Console Purpose

AI Market Twin Admin Console is an internal operations dashboard for managing customers, workspaces, simulations, billing, model versions, usage, system health, and support issues.

Primary users:

- Super Admin
- Operations Manager
- Customer Success Manager
- AI/ML Operations Manager
- Finance/Billing Admin
- Support Agent

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2. Admin Navigation Structure

Left Sidebar Menu

1. Overview
2. Customers
3. Workspaces
4. Projects
5. Simulations
6. Reports
7. Billing
8. Model Management
9. Usage Analytics
10. System Health
11. Support Tickets
12. Audit Logs
13. Admin Settings

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3. Admin Role Permissions

Super Admin

- Full access
- User impersonation
- Billing changes
- Model rollout
- Workspace suspension

Operations Manager

- Customer/workspace/project management
- Simulation monitoring
- Support tools

Customer Success

- Customer view
- Usage analytics
- Reports
- Support tickets

Finance Admin

- Billing
- Invoices
- Plan changes
- Payment failures

ML Operations

- Model versions
- Simulation quality
- Prompt/template management
- Job failure monitoring

Support Agent

- Read customer data
- View logs
- Create support notes
- Limited impersonation if approved

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4. Screen 1 — Admin Overview Dashboard

Purpose

Provide a real-time health snapshot of the business and platform.

Top KPI Cards

- Total Workspaces
- Active Paying Customers
- Monthly Recurring Revenue
- Simulations This Month
- Failed Jobs
- API Usage
- Churn Risk Accounts
- Open Support Tickets

Main Sections

Revenue Snapshot

- MRR
- ARR
- Trial-to-paid conversion
- Payment failures

Platform Health

- Queue length
- Average simulation time
- Job failure rate
- API error rate

Customer Activity

- Top active workspaces
- Dormant accounts
- Quota warnings

CTA Buttons

- View Failed Jobs
- Review Churn Risk
- Export Admin Report

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5. Screen 2 — Customer Management

Purpose

Search, review, and manage all customer accounts.

Table Columns

- Customer Name
- Workspace Name
- Plan
- MRR
- Owner Email
- Status
- Last Active
- Simulations Used
- Created Date

Filters

- Plan
- Status
- Country
- MRR Range
- Last Active
- Trial / Paid / Churned

Row Actions

- View Customer
- Impersonate User
- Change Plan
- Suspend Workspace
- Add Internal Note

Customer Detail Page

Sections

- Company Profile
- Workspace Members

- Subscription Info
- Usage Summary
- Recent Projects
- Simulation History
- Reports Generated
- Support Tickets
- Internal Notes
- Audit History

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6. Screen 3 — Workspace Management

Purpose

Manage multi-tenant workspaces and access controls.

Workspace Detail Fields

- Workspace ID
- Company Name
- Industry
- Country
- Plan
- Owner
- Team Members
- Storage Used
- Simulation Quota
- API Access
- Status

Admin Actions

- Add/remove member
- Change user role
- Reset owner
- Increase quota manually
- Disable exports
- Suspend / reactivate workspace

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7. Screen 4 — Project Management

Purpose

View all customer projects and debug project-level issues.

Table Columns

- Project Name
- Workspace
- Product Category
- Status
- Countries Tested
- Last Updated
- Simulation Count
- Created By

Project Detail Sections

- Product Input Data
- Uploaded Assets
- Candidate Countries
- Competitors
- Simulation Runs
- Generated Reports
- Validation Errors

Admin Actions

- Mark as reviewed
- Re-run validation
- Delete corrupted asset
- Trigger re-simulation
- Export project diagnostic file

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8. Screen 5 — Simulation Operations

Purpose

Monitor and control simulation jobs.

Simulation Queue View

KPI Cards

- Running Jobs
- Queued Jobs
- Completed Today
- Failed Today
- Average Runtime
- Retry Count

Table Columns

- Simulation ID
- Workspace
- Project
- Status
- Model Version
- Persona Count
- Started At
- Runtime
- Error Code

Status Types

- Pending
- Running
- Completed
- Failed
- Cancelled
- Retrying

Admin Actions

- Cancel job
- Retry job
- View logs
- Download raw output
- Mark as false failure
- Escalate to ML Ops

Job Detail Page

Sections

- Job metadata
- Input snapshot
- Stage timeline
- Worker logs
- API/model calls
- Error traces
- Output artifacts

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9. Screen 6 — Report Management

Purpose

Track generated reports and export activity.

Table Columns

- Report Name
- Workspace
- Simulation ID
- Format
- Created By
- Created At
- Download Count
- Share Link Status

Admin Actions

- Rebuild report
- Disable share link
- Download report
- View report preview

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10. Screen 7 — Billing Admin

Purpose

Manage subscriptions, invoices, failed payments, and enterprise billing.

KPI Cards

- MRR
- ARR
- Failed Payments
- Trial Accounts
- Expiring Trials
- Churned Accounts

Billing Table Columns

- Customer
- Plan
- Billing Cycle
- MRR
- Payment Status
- Next Billing Date
- Invoice Status

Admin Actions

- Change plan
- Apply coupon
- Extend trial
- Issue credit
- Cancel subscription
- Mark invoice paid manually
- Generate invoice

Enterprise Billing Section

- Contract start/end date
- Custom quota
- Custom pricing
- PO number
- Invoice contact

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11. Screen 8 — Model Management

Purpose

Control AI model versions, prompt templates, scoring models, and rollout strategy.

Model Registry Table

- Model Version
- Type
- Provider
- Release Status
- Accuracy Score
- Current Traffic %
- Created Date
- Owner

Model Types

- Persona Generator
- Purchase Intent Scorer
- Country Ranking Model
- Pricing Optimizer
- Creative Scoring Model
- Risk Engine
- Report Writer

Admin Actions

- Promote model
- Roll back model
- A/B test model
- Adjust traffic split
- View evaluation results
- Lock model for enterprise customer

Prompt Template Manager

Fields:

- Template name
- Version
- System prompt
- Input variables
- Output schema
- Status
- Last edited by

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12. Screen 9 — Usage Analytics

Purpose

Analyze customer behavior and product usage.

Dashboard Sections

Usage Trends

- Simulations per day
- Reports exported
- Active users
- Country tests by region

Feature Adoption

- Pricing optimizer usage
- Creative test usage
- PDF export usage
- Team invite usage

Customer Health Score

Inputs:

- Login frequency
- Simulation count
- Report export count
- Billing status
- Support ticket frequency

Outputs

- Healthy
- Expansion opportunity
- Churn risk
- Inactive

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13. Screen 10 — System Health

Purpose

Monitor platform reliability and technical performance.

KPI Cards

- API Uptime
- Error Rate
- Queue Delay
- Worker Utilization
- Database Latency
- Storage Usage

Charts

- API response time
- Simulation runtime distribution
- Error rate over time
- Queue depth

Alerts

- High failure rate
- Slow simulations
- Payment webhook failure
- Storage threshold
- Model provider outage

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14. Screen 11 — Support Tickets

Purpose

Help support team resolve customer issues quickly.

Ticket Table Columns

- Ticket ID
- Customer
- Priority
- Category
- Status
- Assigned To
- Created At
- SLA Deadline

Ticket Categories

- Login issue
- Billing issue
- Simulation failed
- Report export issue
- Data accuracy concern
- Enterprise request

Ticket Detail View

Sections:

- Customer info
- Conversation history
- Related project
- Related simulation
- Error logs
- Internal notes
- Resolution status

Actions

- Assign agent
- Change priority

- Add internal note
- Link simulation
- Escalate to engineering
- Close ticket

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15. Screen 12 — Audit Logs

Purpose

Track sensitive admin and customer actions.

Log Fields

- Timestamp
- Actor
- Actor Role
- Workspace
- Action Type
- Resource Type
- Resource ID
- IP Address
- Result

Tracked Actions

- Login
- Impersonation
- Role changes
- Billing changes
- Workspace suspension
- Simulation retry/cancel
- Model rollout
- Report share link disabled

Filters

- Actor
- Workspace
- Action Type
- Date Range

- Result

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16. Screen 13 — Admin Settings

Sections

Admin Users

- Invite admin
- Assign role
- Remove admin

Platform Settings

- Default simulation quota
- Default trial period
- Export limits
- File upload limits

Feature Flags

- Enable video simulation
- Enable PPT export
- Enable API access
- Enable enterprise SSO

Compliance Settings

- Data retention period
- Raw persona storage policy
- Export watermark setting

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17. Critical Security Requirements

Impersonation Rules

- Super Admin only by default
- Must require reason input
- Must be audit logged
- Visible banner during impersonation
- Cannot access billing payment details during impersonation

Billing Protection

- Only Finance Admin and Super Admin can change plans
- Payment method details never exposed

Data Access

- Customer-uploaded assets should use signed URLs
- Internal download actions logged
- Sensitive data access requires elevated permission

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18. Admin Console UI Style

Visual Direction

- Dark sidebar
- White content cards
- Dense but readable tables
- Clear status badges
- Strong search and filters
- Operational command center feel

Status Badge Colors

- Active: Green
- Trial: Blue
- Suspended: Red
- Failed: Red
- Running: Purple
- Queued: Amber
- Completed: Green

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19. Developer Notes

Backend Requirements

- All admin APIs must require admin role verification
- Every sensitive action must write to audit_logs
- Admin table queries need pagination

- Exportable CSV for major tables
- Search must support workspace, email, project, simulation ID

Suggested Admin API Routes

GET /admin/overview GET /admin/customers GET /admin/customers/:id PATCH /admin/workspaces/:id/status POST /admin/workspaces/:id/quota GET /admin/simulations GET /admin/simulations/:id POST /admin/simulations/:id/retry POST /admin/simulations/:id/cancel GET /admin/billing PATCH /admin/billing/:workspaceId/plan GET /admin/models POST /admin/models/:id/promote POST /admin/models/:id/rollback GET /admin/audit-logs GET /admin/system-health

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20. MVP Admin Scope

Must-Have for v0.1

- Admin Overview
- Customer list/detail
- Workspace status management
- Simulation queue monitoring
- Retry failed simulation
- Billing status view
- Audit logs

v0.2

- Model management
- Support tickets
- Usage health score
- Feature flags

v1.0

- Full enterprise billing
- Advanced model rollout
- SLA monitoring
- Automated anomaly detection

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21. Final Admin Console Goal

The admin console should allow the internal team to operate AI Market Twin like a scalable enterprise SaaS business:

- Understand customer health
- Fix simulation issues quickly
- Manage billing safely
- Control model quality
- Monitor system reliability
- Protect customer trust through strict auditability